

COURSE TITLE : INDUSTRIAL INSTRUMENTS I
COURSE CODE : 4213
COURSE CATEGORY : A
PERIODS/WEEK : 4
PERIODS/SEMESTER: 56
CREDITS : 4

TIME SCHEDULE

Module	Topics	Period
1	Pressure measurement	14
2	Level measurement	14
3	Non Electrical type Temperature measurement	14
4	Electrical type Temperature measurement	14
TOTAL		56

COURSE OUTCOME

Module	G.O.	On completion of the study of this module the student will be able
1	1	To know the basics of pressure measurement
	2	To understand the working of different types of manometers
	3	To comprehend the working of elastic type pressure gauges
	4	To understand the working of electrical type pressure gauges
	5	To know the working of vacuum gauges
2	1	To understand the direct level measurement method
	2	To understand the indirect level measurement method
3	1	To know the basics of temperature measurement
	2	To comprehend filled system thermometers
	3	To Understand working of pyrometers
4	1	To Understand Resistance Temperature detectors
	2	To know about Thermistors
	3	To comprehend Thermocouple

On completion of the study the student will be able

MODULE I PRESSURE MEASUREMENT

1.1.0 To Know the basics of pressure measurement

- 1.1.1 To define pressure
- 1.1.2 To define pressure head
- 1.1.3 To compare different types of pressure
- 1.1.4 To list the units of pressure and its conversions

1.2.0 To Understand the working of different types of manometers

- 1.2.1 To classify manometers
- 1.2.2 To explain the working of U tube manometers
- 1.2.3 To describe the working of well type manometers
- 1.2.4 To explain the working of inclined manometers
- 1.2.5 To explain the working of bell type manometers
- 1.2.6 To list the errors in manometers

1.3.0 To Comprehend the working of elastic type pressure gauges

- 1.3.1 To explain the constructional details and working of c-type, spiral and helical bourdon gauges
- 1.3.2 To illustrate the working of bellows
- 1.3.3 To describe the working of diaphragms and capsules
- 1.3.4 To explain the calibration of pressure gauge using Dead weight Tester

1.4.0 To Understand the working of electrical type pressure gauges

- 1.4.1 To explain pressure measurement using strain gauges
- 1.4.2 To describe capacitive type pressure gauge
- 1.4.3 To explain the working of differential pressure transmitter for pressure measurement
- 1.4.4 To explain piezo electric pressure sensor
- 1.4.5 To illustrate fiber optic pressure sensor

1.5.0 To Know the working of vacuum gauges

- 1.5.1 To describe the working of McLeod gauge
- 1.5.2 To explain Pirani gauge
- 1.5.3 To illustrate ionization gauge

MODULE II LEVEL MEASUREMENT

2.1.0 To Understand the working of direct level measurement method

- 2.1.1 To explain sight glass technique
- 2.1.2 To describe float type level indicator method
- 2.1.3 To illustrate the working of displacer and torque tube type level indicator

2.2.0 To Understand the working of indirect level measurement method

- 2.2.1 To explain the working of air purge type level indicator
- 2.2.2 To describe the operation of capacitive level indicator
- 2.2.3 To illustrate the operation of conductive level indicator
- 2.2.4 To explain the working of ultrasonic level gauge

- 2.2.5 To explain the method of level measurement using strain gauges
- 2.2.6 To describe the working of capacitance type level indicator
- 2.2.7 To illustrate the principle of radiation absorption method
- 2.2.8 To explain the laser method used for level indication
- 2.2.9 To explain fiber optic level sensor
- 2.2.10 To describe level switch
- 2.2.11 To explain the working of a level transmitter- Differential Pressure type
- 2.2.12 To explain the calibration of level detectors

MODULE III NON ELECTRICAL TYPE TEMPERATURE MEASUREMENT

3.1.0 To Know the Basics of Temperature Measurement

- 3.1.1 To define temperature
- 3.1.2 To describe different temperature scales and conversions

3.2.0 To Comprehend filled system thermometers

- 3.2.1 To explain the working of bimetallic thermometer
- 3.2.2 To illustrate the working of mercury in glass thermometer
- 3.2.3 To describe the working of mercury in steel thermometer
- 3.2.4 To explain the working of gas and vapour pressure thermometer

3.3.0 To Understand working of pyrometers

- 3.3.1 To explain the construction and working of radiation pyrometer
- 3.3.2 To describe the construction and working of optical pyrometer

MODULE IV ELECTRICAL TYPE TEMPERATURE MEASUREMENT

4.1.0 To Understand Resistance Temperature detectors

- 4.1.1 To explain PTC & NTC
- 4.1.2 To explain Resistance Temperature Detector & Its types (Ni, Cu, & Pt)
- 4.1.3 To explain the characteristics of RTD

4.2.0 To Know Thermistors

- 4.2.1 To describe Thermistor & Its types
- 4.2.2 To explain Thermistor Characteristics
- 4.2.3 To list the applications of Thermistor

4.3.0 To Comprehend Thermocouple

- 4.3.1 To State Seebeck effect, Peltier effect and Thomson effect
- 4.3.2 To State the laws of thermocouple – law of intermediate temperature and law of intermediate metals
- 4.3.3 To list the different types of industrial thermocouples- J, T, K, S, R
- 4.3.4 To explain the properties of different types
- 4.3.5 To describe thermopile
- 4.3.6 To compare the characteristics of Thermocouple, RTD and Thermistor

CONTENT DETAILS

MODULE I

Definition of pressure - pressure head- different types of pressure - units of pressure and its conversions -Classification of manometers - working of U tube manometers, well type manometers , inclined manometers, bell type manometers , errors in manometers - Constructional details and working of C-type , spiral and helical bourdon gauges - Bellows - Diaphragms and capsules - Calibration of pressure gauge using Dead weight Tester - Strain gauges- Capacitive type pressure gauge- piezo electric pressure sensor- Fiber optic pressure sensor - Differential pressure transmitter for pressure measurement - Construction and working of McLeod gauge, Pirani gauge - Ionization gauge

MODULE II

Sight glass technique - Float type level indicator method- Displacer and torque tube type level indicator. Air purge type level indicator - Capacitive and conductive level indicator - Ultrasonic level gauge - Strain gauges - Capacitance type level indicator - Radiation absorption method and laser method used for level indication- fiber optic level sensor - Level switch – Differential Pressure type level transmitter - Calibration of level detectors.

MODULE III

Definition of temperature - different temperature scales and conversions - Working of bimetallic thermometer , Mercury in glass thermometer, Mercury in steel thermometer, Gas and vapour pressure thermometer - Construction and working of radiation pyrometer, Optical pyrometer.

MODULE IV

Definition of PTC & NTC – construction and working of Resistance Temperature Detector – different types (Ni, Cu, & Pt) - characteristics of RTD - Thermistor & Its types- properties- Thermistor Characteristics- Applications of Thermistor - Seebeck effect - Peltier effect and Thomson effect - laws of thermocouple – law of intermediate temperature and law of intermediate metals – different types of industrial thermocouples- J, T, K, S, R - properties of different types - thermopile - Compare the characteristics of Thermocouple, RTD and Thermistor.

REFERENCES

1. A.K.Sawhney, A course in Mechanical measurements and Instrumentation, Dhanpat Rai
2. S.K Singh , Industrial Instrumentation and Control , TMH
3. R.K Jain , Mechanical and Industrial measurements , Khanna Publishers
4. Krishnaswami , Industrial instrumentation , New Age Publication